

Research Question 1: Effectiveness of discharge/care-transition interventions on 30-day readmission

- **Treatments:** AnchorStayMedsPriorToDC, AnchorStayTOCConsult, TCMVisitDate (or “TCM within X days”), TCMClinicianType
 - AnchorStayTOCConsult will be our main interaction term for this research question and the rest of the model
- **Outcome Variable:** 30DayReadmissionFlag
- **What we are expecting/want to see:**
 - AnchorStayMedsPriorToDC=1 → lower 30-day readmission odds
 - AnchorStayTOCConsult → lower 30-day readmission odds
 - TCM by Physician vs APP differs in the association with readmission

Research Question 2: Heterogeneous effectiveness (who benefits most from these types of programs)

- **Moderators:** various socio-demographic and risk factors
- **What are we expecting/want to see:**
 - Effects are stronger for higher risk patients
 - Risk is referring to readmission risk
 - Effects will differ across MDCCategory
 - Major disease category is the primary reason for the index stay hospitalization

1. Data Prep and Feature Engineering

- a. Change column names to be consistent
- b. Time Windows for TCM
 - i. Create TCM_7d, TCM_10d, TCM_14d, and TCM_30d variables
- c. Stratify/encode MDCCategory as a factor and AgeGroup as an ordered factor
 - i. PN = 1, HF = 2, etc.
 - ii. Ordering age group so that 18-39 < 40-49 < 50-59 < 60-69 < 70-79 < 80+
- d. Composite “Transition Bundle” exposure (TOC_bundle = MedsPriorToDC + TOCConsult + TCM_7d)
 - i. Potentially do 10d and 14d if we have time
- e. Missingness Rules:
 - i. DaysBetweenDCAndReadmission is missing when no hospitalization occurs
 1. Do not credit to the person, instead censor the cases where there is no readmission
 - ii. Blank TCMVisitDate = no observed TCM at UHS or at all

1. In the main models, run this as “no TCM recorded and then run the given sensitivity analysis

2. Descriptive and Balance Diagnostics

- a. Generate Readmission rate
 - i. Use MDCCategory, ReadmissionRiskCategory, AgeGroup, Social Determinants of Health (SODH, last columns in the dataset)
- b. Treatment Prevalence
 - i. How much do these treatments affect the readmission rate and diagnostics
 1. Meds, Consult, TCM (given in research question 1)
- c. “Balance” Checks:
 - i. Compare “risk” score (aka risk value) and SDOH prevalence between treated patients and untreated patients in the above categories
 - ii. Standardise mean difference pre-and post-weighting
 1. Compare the difference in means between two groups of units of standard deviation, usually $(X1 - X2) / (\text{Pooled Standard Deviation})$
 - a. 0 means no difference in the means
 - b. Pre - covariate differences in raw data (usually high)
 - c. Post - want to see a significant reduction in the SMD
 - d. Will be a range between 0-1, want to see < 0.1 for good covariate balance
 2. Ex. age in years and blood pressure in mmHg

Deliverable 1: Descriptive Statistics table (overall + by treatment)

3. Main Analysis

3.1 Primary Model - 30-Day readmission

Create a logit model with controls and matching to answer the following question: Among otherwise similar patients, do the patients that received the intervention have different outcomes than the patients that did not receive intervention?

$\text{logit}(P(\text{Readmit30} = 1)) = \beta_0 + \text{meds} + \text{TOCConsult} + \text{TCM7d} + f(\text{RiskScore}) + \text{MDC} + \text{Covariates}$

- $f(\text{Risk Score}) \rightarrow$ Function of Patient risk score, generated in step 2
- MDC $\rightarrow$ MDCCategory (why were they admitted to the hospital)
- Covariates

Recommended Covariate Set:

- Demographics

- Health behaviors (smokingstatus, tobacco and alcohol use)
- SDOH Flags

Notes:

- Use robust standard errors cluster at the patient level
 - This address the within subject correlation (multiple measurements per patient)
 - Does not change point estimates but adjusts the standard errors instead
 - Expect to see wider confidence intervals
- Report Marginal effects (AME) alongside odds ratios for interpretability
 - Risk insights
 - Average percentage point change in outcome per unit change in a predictor
 - OR → Relative odds

3.2 Treatment Bundle and Dose Response

Replace separate interventions with TOC_Bundle (0-3):

$$\text{logit}(P(\text{Readmit30} = 1)) = \beta_0 + \sum_{k=1}^3 \beta_k 1\{\text{bundle} = k\} + \text{controls}$$

Tests additive benefit of multi-component transitions

3.3 Herogeneity and Moderation

Add interactions to the model

Ex.

1. TCM7d × ReadmissionRiskValue (or risk category)
2. TOCConsult × HealthLiteracy (or Transportation Needs)
3. TCM7d × MDCCategory (selective, avoid overfitting)

This will take lots of trial and error to create helpful interactions

Other Tasks Before Mid-March:

1. Strings need to be encoded into category numbers
2. Z-score, log transformation, etc.
3. Change up the Column names to be consistent
4. Percentile changes (remove extreme values)
5. Encode AgeGroup, MDCCategory, MaritalStatus, SmokingStatus, SexualOrientation (maybe)